Cyclic groups and counting matrices

Multiplication in a finite field has a generator

$$K^\times = \{1, g, g^2, \ldots, g^{q-2}\}, \qquad |K| = q.$$

The nonzero elements form a group

Let K be a field with q elements.

$$K^\times = K \setminus \{0\}, \qquad |K^\times| = N = q - 1.$$

Multiplication is commutative; the identity is 1; every element has an inverse.

$$\mathrm{ord}(a) \mid N, \qquad a^N = 1 \quad (a \in K^\times).$$

What cyclic means

In $\mathbb{F}_5$, the powers of 2 are

t	0	1	2	3
2^t	1	2	4	3

$$2^4 = 1.$$

Every nonzero element is a power of 2.

A group is cyclic if one element generates the whole group.

A polynomial cannot have too many roots

Over a field, if $f(a) = 0$, then

$$f(X) = (X - a)h(X).$$

Any other root $b \neq a$ is a root of h:

$$0 = (b - a)h(b) \quad \Longrightarrow \quad h(b) = 0.$$

Induction gives:

$$\deg f = d, \ f \neq 0 \quad \Longrightarrow \quad \text{at most } d \text{ roots.}$$

Build the order one prime at a time

Factor $N = q - 1$:

$$N = \ell_1^{r_1} \cdots \ell_t^{r_t}.$$

We will construct, for each prime factor ℓ,

$$g_\ell \text{ with order } \ell^r, \qquad \ell^r \parallel N.$$

Then multiply these elements.

If $N = 1$, the group $\{1\}$ is already cyclic.

Find an element with the required prime factor

Fix $\ell^r \parallel N$.

The polynomial

$$X^{N/\ell} - 1$$

has at most $N/\ell < N$ roots.

It cannot vanish at every element of $K^\times$.

Choose $a \in K^\times$ with $a^{N/\ell} \neq 1$.

Isolate the prime-power order

Set $M = N/\ell^r$ and $g_\ell = a^M$.

$$g_\ell^{\ell^r} = a^N = 1, \qquad g_\ell^{\ell^{r-1}} = a^{N/\ell} \neq 1.$$

Its order divides ℓ^r, but does not divide ℓ^{r-1}.

Every proper divisor of ℓ^r divides ℓ^{r-1}.

$$\operatorname{ord}(g_\ell) = \ell^r.$$

Combine two coprime orders

Suppose a, b commute, with orders r, s and $\gcd(r, s) = 1$.

$$(ab)^{rs} = 1.$$

If $(ab)^t = 1$, put $z = a^t = b^{-t}$. Then

$$z^r = z^s = 1 \quad \Longrightarrow \quad z = 1$$

by Bezout's identity. Thus $r \mid t$ and $s \mid t$, so $rs \mid t$.

$$\operatorname{ord}(ab) = rs.$$

One element of full order

The prime-power orders $\ell_i^{r_i}$ are pairwise coprime.

By repeatedly applying the lemma,

$$g = g_{\ell_1} \cdots g_{\ell_t} \quad \text{has order} \quad \ell_1^{r_1} \cdots \ell_t^{r_t} = N.$$

Its N distinct powers exhaust $K^\times$.

The multiplicative group of every finite field is cyclic.

Primitive elements

A generator g of $K^\times$ is called a primitive element.

$$K^\times = \{1, g, \ldots, g^{q-2}\}, \qquad \operatorname{ord}(g) = q - 1.$$

If K has characteristic p, then

$$K = \mathbb{F}_p(g) : \text{ the field is generated by one element.}$$

Every nonzero element is a power of g.

Your turn: combine orders in $\mathbb{F}_7$

In $\mathbb{F}_7^\times$,

$$\operatorname{ord}(6) = 2, \qquad \operatorname{ord}(4) = 3.$$

Verify these orders, then answer:

1. What is the order of $6 \cdot 4$?
2. Which residue is therefore a primitive element?

A generator modulo seven

$$6 \cdot 4 = 3 \pmod 7, \qquad \operatorname{ord}(3) = 2 \cdot 3 = 6.$$

t	0	1	2	3	4	5
3^t	1	3	2	6	4	5

One element generates all six nonzero residues.

From field elements to matrices

Over $\mathbb{F}_p$, the span of r independent vectors has p^r elements.

An invertible matrix has columns forming an ordered basis.

Count invertible matrices by choosing their columns.

Invertible matrices and ordered bases

For a matrix with columns $v_1, \ldots, v_d$,

$$M \begin{pmatrix} a_1 \\ \vdots \\ a_d \end{pmatrix} = a_1 v_1 + \cdots + a_d v_d.$$

M is invertible exactly when every output has unique coordinates.

Its columns must form an ordered basis of $\mathbb{F}_p^d$.

Choose the first column

In $\mathbb{F}_p^2$, there are p^2 vectors.

The first basis vector can be anything except zero.

$p^2 - 1$ choices.

$$v_1 \neq \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

Choose the second column

Once $v_1 \neq 0$ is fixed, its span has p elements:

$$\operatorname{span}(v_1) = \{av_1 : a \in \mathbb{F}_p\}.$$

Choose v_2 outside that span:

$$p^2 - p \text{ choices.}$$

Therefore

$$|\operatorname{GL}_2(\mathbb{F}_p)| = (p^2 - 1)(p^2 - p).$$

Your turn: count modulo two

How many invertible 2×2 matrices are there over $\mathbb{F}_2$?

Fix the first column to be

$$v_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

Which second columns are allowed?

Six invertible matrices

$$|\,\mathrm{GL}_2(\mathbb{F}_2)| = (4-1)(4-2) = 6.$$

With $v_1 = (1,0)^{\mathsf{T}}$, the allowed second columns are

$$\begin{pmatrix} 0 \\ 1 \end{pmatrix}, \qquad \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

The span we excluded is $\{(0,0),(1,0)\}$.

The same count in dimension d

After r independent columns, the span has p^r vectors.

The next column has $p^d - p^r$ choices.

$$|\operatorname{GL}_d(\mathbb{F}_p)| = \prod_{r=0}^{d-1} (p^d - p^r).$$

The first factor is $p^d - 1$; the last is $p^d - p^{d-1}$.

The determinant groups matrices by value

$$\det : \mathrm{GL}_d(\mathbb{F}_p) \longrightarrow \mathbb{F}_p^\times.$$

$$\det(MN) = \det(M)\det(N).$$

Every nonzero $a \in \mathbb{F}_p$ occurs:

$$D_a = \mathrm{diag}(a, 1, \ldots, 1), \qquad \det D_a = a.$$

$\mathrm{SL}_d(\mathbb{F}_p)$ is the set with determinant 1.

Every determinant fiber has the same size

Multiplication by D_a gives a bijection

$$\mathrm{SL}_d(\mathbb{F}_p) \longrightarrow \{M : \det M = a\}, \qquad S \longmapsto D_a S.$$

The inverse is $M \longmapsto D_a^{-1} M$.

There are $p-1$ nonzero determinant values, so

$$|\mathrm{SL}_d(\mathbb{F}_p)| = \frac{|\mathrm{GL}_d(\mathbb{F}_p)|}{p-1}.$$

In dimension two

$$|\operatorname{SL}_2(\mathbb{F}_p)| = \frac{(p^2-1)(p^2-p)}{p-1}$$
$$= p(p^2-1).$$

Lagrange's theorem: the order of a group element divides the group size.

Next time: apply this count to Fibonacci.

Exit question

For a field K with q elements, which group have we proved is cyclic?

What is its order?

Why does this not say that every nonzero element generates it?

The determinant quotient

The determinant is surjective, with kernel $\mathrm{SL}_d(\mathbb{F}_p)$.

$$\mathrm{GL}_d(\mathbb{F}_p)/\mathrm{SL}_d(\mathbb{F}_p) \cong \mathbb{F}_p^\times.$$

A coset is determined by the common determinant of its matrices.

Multiplication of cosets corresponds to multiplication of determinants.

A different object with a cyclic unit group

For odd primes p and $n \geq 1$,

$$(\mathbb{Z}/p^n\mathbb{Z})^\times \text{ is cyclic of order } p^{n-1}(p-1).$$

For $n > 1$, the ring is not a field: $p \cdot p^{n-1} = 0$.

	$\mathbb{F}_{p^n}$	$\mathbb{Z}/p^n\mathbb{Z}$
Units	$p^n - 1$	$p^{n-1}(p-1)$
Characteristic	p	p^n

Two arithmetics with nine elements

In $\mathbb{Z}/9\mathbb{Z}$, the powers of 2 are

$$1, 2, 4, 8, 7, 5, \qquad 2^6 = 1.$$

These are all six units; the nonzero residues $3, 6$ are not units.

In $\mathbb{F}_9$, all eight nonzero elements are units, and a primitive element has order 8.

First lift a primitive root to modulo p^2

Let p be odd, and let g generate $\mathbb{F}_p^\times$.

Choose $h = g$ or $h = g + p$ so that

$$h^{p-1} \not\equiv 1 \pmod{p^2}.$$

This is possible because

$$(g+p)^{p-1} \equiv g^{p-1} + p(p-1)g^{p-2} \pmod{p^2},$$

and the added term is nonzero modulo p^2.

Each further lift contributes a factor of p

For odd p, if $u = 1 + p^r c$ with $r \geq 1$ and $p \nmid c$, then

$$u^p = 1 + p^{r+1} c', \qquad p \nmid c'.$$

The first binomial term is $p^{r+1}c$; all later terms are divisible by p^{r+2}.

Starting with $h^{p-1} = 1 + pc$ gives

$$v_p\left(h^{(p-1)p^k} - 1\right) = k + 1.$$

The lifted element has full order

Modulo p^n, let $d = \mathrm{ord}(h)$.

$$p - 1 \mid d, \qquad d \mid (p-1)p^{n-1}.$$

Thus $d = (p-1)p^j$ for some $0 \leq j \leq n-1$.

The lifting calculation shows the least such j is $n-1$.

$$d = (p-1)p^{n-1}.$$

So h generates every unit modulo p^n.

The exception starts at eight

$$(\mathbb{Z}/2\mathbb{Z})^\times = \{1\}, \qquad (\mathbb{Z}/4\mathbb{Z})^\times = \{1, 3\}.$$

Both are cyclic. Modulo 8, every unit squares to 1:

$$1^2 = 3^2 = 5^2 = 7^2 = 1 \pmod 8.$$

There is no element of order 4.

For $n \geq 3$,

$$(\mathbb{Z}/2^n\mathbb{Z})^\times \cong C_2 \times C_{2^{n-2}}.$$

Why the field proof does not apply to residue rings

In $\mathbb{Z}/8\mathbb{Z}$, the degree-two polynomial

$$X^2 - 1$$

has four roots: $1, 3, 5, 7$.

The polynomial root bound required a field.

An abelian unit group need not be cyclic.